



2009 Trial Examination

FORM VI

MATHEMATICS EXTENSION 2

Tuesday 11th August 2009

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Structure of the paper

- Total marks — 120
- All eight questions may be attempted.
- All eight questions are of equal value.

Collection

- Write your candidate number clearly on each booklet and on the tear-off sheet.
- Hand in the eight questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Bundle the tear-off sheet with the question it belongs to.
- Bundle the tear-off sheet with the question it belongs to.
- Place the question paper inside your answer booklet for Question 1.

Checklist

- SGS booklets — 8 per boy
- Candidature — 72 boys

Examiner

DNW

QUESTION ONE (15 marks) Use a separate writing booklet.**Marks**

(a) Evaluate $\int_0^1 x e^{x^2} dx$. **2**

(b) Complete the square to find $\int \frac{dx}{x^2 - 2x + 5}$. **2**

(c) Evaluate $\int_0^{\frac{\pi}{2}} x \sin x dx$. **3**

(d) (i) Find values of a , b and c such that **3**

$$\frac{x^2 + 2x - 4}{(x + 1)(x^2 + 4)} = \frac{a}{x + 1} + \frac{bx + c}{x^2 + 4}.$$

(ii) Hence evaluate $\int_0^1 \frac{x^2 + 2x - 4}{(x + 1)(x^2 + 4)} dx$. **3**

(e) Use the substitution $x = \frac{\pi}{2} - u$ to show that **2**

$$\int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \sin 2x} dx = 0.$$

QUESTION TWO (15 marks) Use a separate writing booklet.**Marks**(a) Let $z = 3 - 4i$ and $w = 2 + i$. Find, in the form $x + iy$:

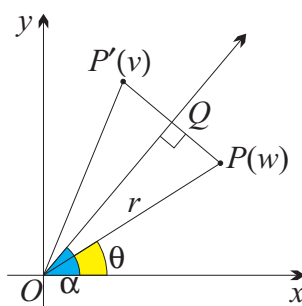
(i) $z + iw$

1

(ii) $z \overline{w}$

1(b) Let $\alpha = 1 - i$.(i) Write α in modulus-argument form.**1**(ii) Hence show that $\alpha^4 + 4 = 0$.**2**(c) Let $z = x + iy$ and $w = 1 - \frac{2i}{z}$.(i) Write w in the form $a + ib$.**2**(ii) For what value of z is w undefined?**1**(iii) Given that w is purely imaginary, describe the locus of z .**2**

(d)



In the Argand diagram above, P represents the complex number $w = r \operatorname{cis} \theta$. Q is that point on the ray $\arg(z) = \alpha$ such that $\angle PQO = \frac{\pi}{2}$. The point P' , which represents the complex number v , is the reflection of P in the ray $\arg(z) = \alpha$. You may assume that $\triangle OPQ \equiv \triangle OP'Q$.

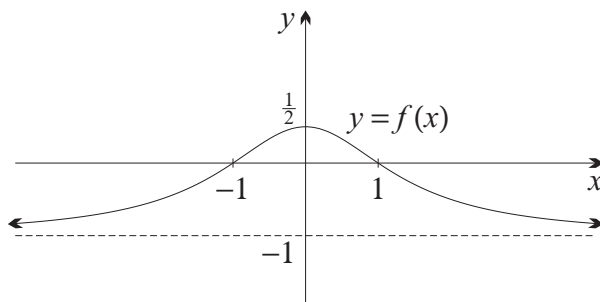
(i) Write down the values of $|v|$ and $\arg(v)$.**2**(ii) Hence show that $v = \overline{w} \operatorname{cis} 2\alpha$.**1**(iii) The circle $|z - (2 + 2i)| = 1$ is reflected in the ray $\arg(z) = \frac{\pi}{6}$. By using the result in part (ii), or otherwise, show that the equation of this new circle is**2**

$$\left| z - ((1 + \sqrt{3}) + i(\sqrt{3} - 1)) \right| = 1.$$

QUESTION THREE (15 marks) Use a separate writing booklet.

Marks

(a)



The graph of $y = f(x)$ is shown above. The horizontal asymptote is $y = -1$ and the y -intercept is at $(0, \frac{1}{2})$. The x -intercepts are at $(-1, 0)$ and $(1, 0)$.

Draw separate graphs of the following functions:

(i) $y = \frac{1}{f(x)}$

2

(ii) $y = (f(x))^2$

2

(iii) $y = 4^{f(x)}$

2

(b) The ellipse $\mathcal{E}$ has equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

(i) State the intercepts with the axes.

1

(ii) Determine the eccentricity of $\mathcal{E}$.

1

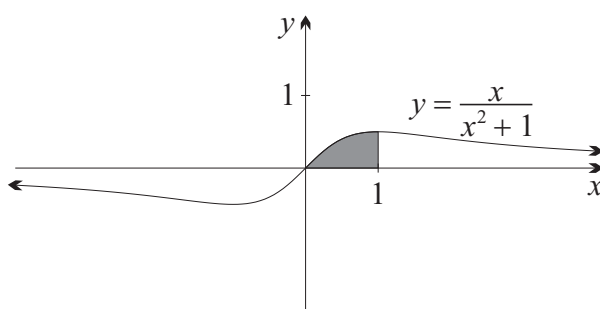
(iii) State the coordinates of the two foci.

1

(iv) Find the equations of the two directrices.

1

(c)



5

The graph of $y = \frac{x}{x^2 + 1}$ is shown above.

Use the method of cylindrical shells to find the volume of the solid generated when the shaded region bounded by $y = 0$, $y = \frac{x}{x^2 + 1}$ and $x = 1$ is rotated about the y -axis.

QUESTION FOUR (15 marks) Use a separate writing booklet.**Marks**(a) (i) Show that $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$.**1**(ii) Hence evaluate $\int_0^{\frac{\pi}{3}} 2 \cos 2x \sin x \, dx$.**2**(b) Consider the integral $I_n = \int_0^1 \frac{x^n}{\sqrt{1+x}} \, dx$.(i) Show that $I_0 = 2\sqrt{2} - 2$.**1**(ii) Show that $I_{n-1} + I_n = \int_0^1 x^{n-1} \sqrt{1+x} \, dx$.**1**

(iii) Use integration by parts to show that

2

$$I_n = \frac{2\sqrt{2} - 2nI_{n-1}}{2n+1}.$$

(iv) Hence evaluate I_2 .**2**(c) The number c is real and non-zero. It is also known that $(1 + ic)^5$ is real.(i) Use the binomial theorem to expand $(1 + ic)^5$.**1**(ii) Show that $c^4 - 10c^2 + 5 = 0$.**2**(iii) Hence show that $c = \sqrt{5 - 2\sqrt{5}}, -\sqrt{5 - 2\sqrt{5}}, \sqrt{5 + 2\sqrt{5}}$ or $-\sqrt{5 + 2\sqrt{5}}$.**1**(iv) Let $1 + ic = r \operatorname{cis} \theta$. Use de Moivre's theorem to show that the smallest positive value of θ is $\frac{\pi}{5}$.**1**(v) Hence evaluate $\tan \frac{\pi}{5}$.**1**

QUESTION FIVE (15 marks) Use a separate writing booklet.

Marks

(a) The polynomial $P(z) = 2z^3 - 3z^2 + 8z + 5$ has a zero at $z = 1 - 2i$. Factorise $P(z)$. **3**

(b) (i) The cubic equation $x^3 - px - q = 0$ has a double root. Show that $27q^2 = 4p^3$. **3**

(ii) Hence find the y -coordinates of the stationary points of $y = x^3 - 3x$ without the use of calculus. **1**

(c) Consider the series:

$$S = 1 - x^2 + x^4 - x^6 + \dots$$

(i) For which values of x does S have a limiting sum, and what is the limiting sum? **2**

(ii) Assuming that it is valid to integrate this series, show that **3**

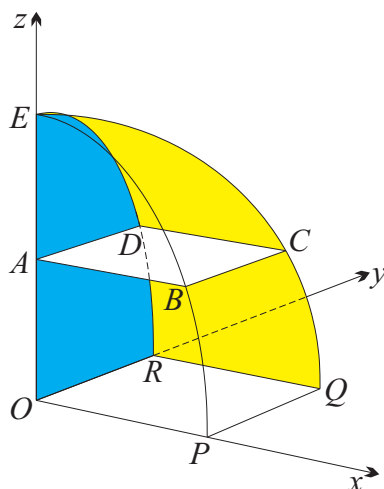
$$\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \dots$$

(iii) Show that $\tan \frac{\pi}{12} = 2 - \sqrt{3}$. **2**

(iv) Let $x = \tan \frac{\pi}{12}$. Use this value of x and the first three terms of the series in part (ii) to find an approximation for π , correct to four decimal places. **1**

QUESTION SIX (15 marks) Use a separate writing booklet.**Marks**

(a)



The solid in the diagram above has a horizontal square base $OPQR$ with diagonal $OQ = r$. The thin horizontal slice $ABCD$ at height z above the base is also square with $OC = r$. The line OA is vertical. The curve QCE is a quadrant of a circle with centre O and radius r .

(i) Show that the area of $ABCD$ is $\frac{1}{2}(r^2 - z^2)$.

2

(ii) Hence find the volume of the solid.

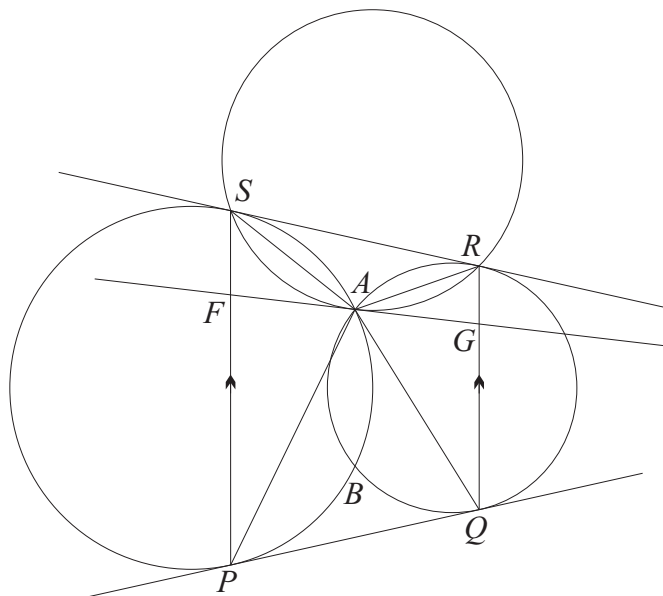
3

(b) The points $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$ lie on the same branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, and PQ is a focal chord, passing through $S(ae, 0)$.

4

Use the gradients of PS and QS to show that $e = \frac{\sin \theta - \sin \phi}{\sin(\theta - \phi)}$.

(c)



In the diagram above, two circles of differing radius intersect at A and B . The lines PQ and RS are the common tangents with $PS \parallel QR$. A third circle passes through the points S , A and R . The tangent to this circle at A meets the parallel lines at F and G .

Let $\angle RAG = \alpha$, $\angle AGR = \beta$ and $\angle GRA = \gamma$.

NOTE: You do not have to copy the diagram above. It has been reproduced for you on a tear-off sheet at the end of this paper. Insert this sheet into your answer booklet.

- (i) State why $\angle AFP = \beta$. 1
- (ii) Show that $\angle SPA = \alpha$. 2
- (iii) Hence prove that FG is also tangent to the circle which passes through the points A , P and Q . 3

QUESTION SEVEN (15 marks) Use a separate writing booklet.**Marks**

- (a) (i) The definition of kC_r is the coefficient of x^r in the expansion of $(1+x)^k$. Using this definition, what is the value of kC_r whenever $k < r$? **1**

- (ii) Prove that $\sum_{k=0}^n {}^kC_r = {}^{n+1}C_{r+1}$. You may assume the addition property for the binomial coefficients, which may be written as ${}^kC_r = {}^{k+1}C_{r+1} - {}^kC_{r+1}$. **2**

- (iii) Use the result proven in part (ii) to show that $\sum_{k=0}^n k = \frac{1}{2}n(n+1)$. **1**

- (iv) (α) Show that $k^2 = 2 \times {}^kC_2 + {}^kC_1$. **1**

- (β) Hence find a formula for $\sum_{k=0}^n k^2$. **2**

- (b) Show that the equation of the directrix of the parabola $y = ax^2 + bx$ is **2**

$$y = -\frac{b^2 + 1}{4a}.$$

- (c) A projectile is fired from the origin O with initial speed V and angle of projection α . The Cartesian equation of its trajectory is

$$y = x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2V^2}.$$

- (i) Use part (b) to find the equation of the directrix. **2**

- (ii) Hence show that the focus lies on the circle **1**

$$x^2 + y^2 = \frac{V^4}{4g^2}.$$

- (iii) There is only one trajectory which passes through P . Use the geometry of the parabola to prove that OP is a focal chord. **3**

QUESTION EIGHT (15 marks) Use a separate writing booklet.**Marks**

- (a) Consider the function

$$f(x) = x - \frac{g^2}{x} - 2g \log \left(\frac{x}{g} \right), \text{ for } x \geq g.$$

- (i) Evaluate
- $f(g)$
- .

1

- (ii) Show that
- $f'(x) = \left(1 - \frac{g}{x}\right)^2$
- .

1

- (iii) Explain why
- $f(x) > 0$
- for
- $x > g$
- .

1

- (b) A body is moving vertically through a resisting medium, with resistance proportional to its speed. The body is initially fired upwards from the origin with speed
- v_0
- . Let
- y
- metres be the height of the object above the origin at time
- t
- seconds, and let
- g
- be the constant acceleration due to gravity. Thus

$$\frac{d^2y}{dt^2} = -(g + kv) \quad \text{where } k > 0.$$

- (i) Find
- v
- as a function of
- t
- , and hence show that

4

$$k^2y = (g + kv_0)(1 - e^{-kt}) - gkt.$$

- (ii) Find
- T
- , the time taken to reach the maximum height.

1

- (iii) Show that when
- $t = 2T$
- ,

1

$$k^2y = (g + kv_0) - \frac{g^2}{g + kv_0} - 2g \log \left(\frac{g + kv_0}{g} \right).$$

- (iv) Use this result and part (a) to show that the downwards journey takes longer.

1

- (c) Suppose that the equation
- $f(x) = 0$
- has a single root
- $x = \alpha$
- , where
- $a \leq \alpha \leq b$
- . Let the sequence

$$x_0 = a, \quad x_1 = b, \quad x_2 = \frac{a+b}{2}, \quad x_3, \quad x_4, \quad \dots$$

be the successive approximations of $x = \alpha$ obtained when the bisection method is used. (The bisection method is also known as the method of halving the interval.)

Let $u_n = |x_n - x_{n-1}|$ be the distances between successive terms of this sequence.

- (i) Explain why
- $u_{n+1} = \frac{1}{2}u_n$
- .

1

- (ii) Hence show that
- $u_n = (b - a) \left(\frac{1}{2}\right)^{n-1}$
- for
- $n \geq 1$
- .

2

- (iii) Explain why
- $|\alpha - x_n| \leq u_n$
- .

1

- (iv) Hence prove that the bisection method converges to the root
- $x = \alpha$
- .

1

That is, prove that $\lim_{n \rightarrow \infty} x_n = \alpha$.

END OF EXAMINATION**Tear-off pages follow ...**

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

QUESTION ONE (15 marks)

$$\begin{aligned} \text{(a)} \quad \int_0^1 x e^{x^2} dx &= \frac{1}{2} \left[e^{x^2} \right]_0^1 \\ &= \frac{1}{2}(e - 1). \end{aligned}$$

☒
☒

$$\begin{aligned} \text{(b)} \quad \int \frac{dx}{x^2 - 2x + 5} &= \int \frac{dx}{(x - 1)^2 + 2^2} \\ &= \frac{1}{2} \tan^{-1} \frac{x - 1}{2} + C \end{aligned}$$

☒
☒

$$\begin{aligned} \text{(c)} \quad \int_0^{\frac{\pi}{2}} x \sin x dx &= \left[-x \cos x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x dx \quad (\text{by parts}) \\ &= \left[-x \cos x + \sin x \right]_0^{\frac{\pi}{2}} \\ &= 1 \end{aligned}$$

☒
☒
☒

(d) (i) The given equation is true if

$$x^2 + 2x - 4 = a(x^2 + 4) + (bx + c)(x + 1).$$

$$\text{At } x = -1 \quad -5 = 5a + 0$$

$$\text{so} \quad a = -1.$$

$$\text{At } x = 0 \quad -4 = -4 + c$$

$$\text{so} \quad c = 0.$$

$$\text{At } x = 1 \quad -1 = -5 + 2b$$

$$\text{so} \quad b = 2.$$

$$\begin{aligned} \text{(ii)} \quad \int_0^1 \frac{x^2 + 2x - 4}{(x + 1)(x^2 + 4)} dx &= \int_0^1 \frac{2x}{x^2 + 4} - \frac{1}{x + 1} dx \quad (\text{from part (i)}) \\ &= \left[\log(x^2 + 4) - \log(x + 1) \right]_0^1 \\ &= \log 5 - \log 2 - \log 4 + \log 1 \\ &= \log \left(\frac{5}{8} \right) \end{aligned}$$

☒
☒
☒

$$\text{(e) Let} \quad I = \int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \sin 2x} dx$$

$$\text{Put} \quad x = \frac{\pi}{2} - u$$

$$\text{then} \quad dx = -du.$$

$$\text{at} \quad x = 0, \quad u = \frac{\pi}{2}$$

$$\text{and at} \quad x = \frac{\pi}{2}, \quad u = 0.$$

$$\begin{aligned}
 \text{Thus } I &= \int_{\frac{\pi}{2}}^0 \frac{\cos(\frac{\pi}{2} - u) - \sin(\frac{\pi}{2} - u)}{1 + \sin 2(\frac{\pi}{2} - u)} (-du) & \checkmark \\
 &= \int_0^{\frac{\pi}{2}} \frac{\cos(\frac{\pi}{2} - u) - \sin(\frac{\pi}{2} - u)}{1 + \sin 2(\frac{\pi}{2} - u)} du \\
 &= \int_0^{\frac{\pi}{2}} \frac{\sin u - \cos u}{1 + \sin 2u} du \\
 \text{so } I &= -I & \checkmark \\
 \text{Hence } I &= 0
 \end{aligned}$$

Total for Question 1: 15 Marks

QUESTION TWO (15 marks)

$$\begin{aligned}
 \text{(a) (i) } z + iw &= 3 - 4i + 2i - 1 \\
 &= 2 - 2i & \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } z \overline{w} &= (3 - 4i)(2 - i) \\
 &= 2 - 11i & \checkmark
 \end{aligned}$$

$$\text{(b) (i) } 1 - i = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) & \checkmark$$

$$\begin{aligned}
 \text{(ii) } \alpha^4 + 4 &= \left(\sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)\right)^4 + 4 \\
 &= 4 \operatorname{cis}(-\pi) + 4 \quad (\text{by de Moivre}) & \checkmark \\
 &= -4 + 4 & \checkmark \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) (i) } w &= 1 - \frac{2i}{z} \\
 &= 1 - \frac{2i \overline{z}}{|z|^2} & \checkmark \\
 &= \left(1 - \frac{2y}{x^2 + y^2}\right) - \frac{2ix}{x^2 + y^2} & \checkmark
 \end{aligned}$$

$$\text{(ii) } w \text{ is undefined when } z = 0 & \checkmark$$

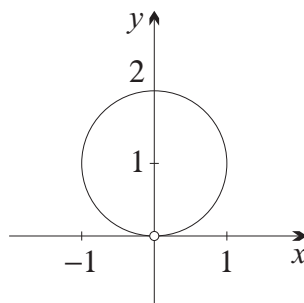
(iii) Since w is pure imaginary,

$$\operatorname{Re}(w) = 0$$

$$\text{so } x^2 + y^2 - 2y = 0$$

$$\text{or } x^2 + (y - 1)^2 = 1$$

Thus the locus is the unit circle with centre $z = i$, omitting the origin.



$\checkmark$

$\checkmark$

(d) (i) $|v| = |w|$ (since $OP' = OP$)

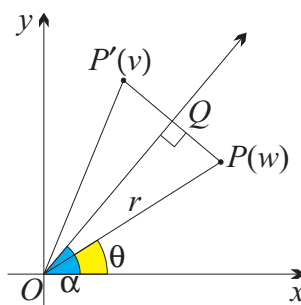
$$= r$$

$$\arg v = \alpha + \angle P'OQ$$

$$= \alpha + \angle POQ$$

$$= \alpha + (\alpha - \theta)$$

$$= 2\alpha - \theta$$



(ii) $v = r \operatorname{cis}(2\alpha - \theta)$

$$= r \operatorname{cis}(-\theta) \operatorname{cis} 2\alpha$$

$$= \overline{w} \operatorname{cis} 2\alpha$$



(iii) The radius remains the same for a reflection. The new centre will be

$$\overline{(2 + 2i)} \operatorname{cis}(2 \times \frac{\pi}{6}) = (2 - 2i) \operatorname{cis} \frac{\pi}{3}$$

$$= (2 - 2i)^{\frac{1}{2}}(1 + i\sqrt{3})$$

$$= (1 + \sqrt{3}) + i(\sqrt{3} - 1)$$



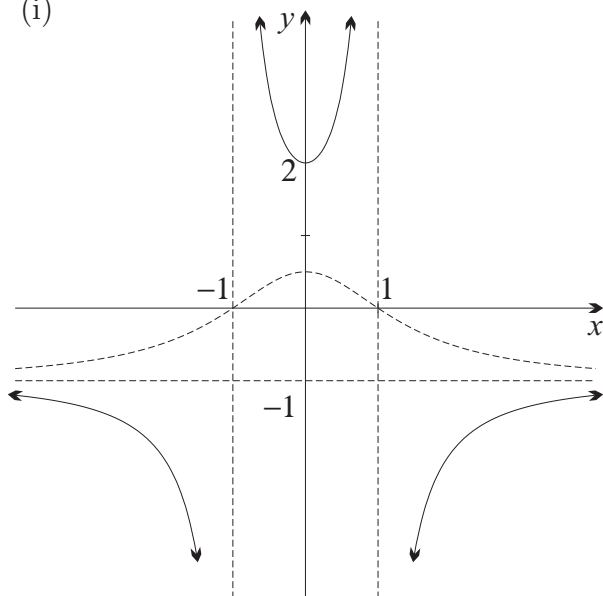
Hence the new circle is $\left| z - ((1 + \sqrt{3}) + i(\sqrt{3} - 1)) \right| = 1$.

Total for Question 2: 15 Marks

QUESTION THREE (15 marks)

(a) The graphs below are exact.

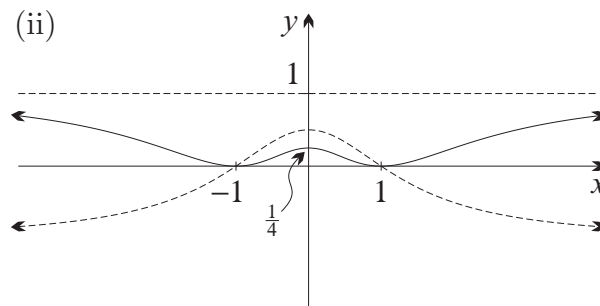
(i)



vertical asymptotes

y-intercept and horizontal asymptote

(ii)

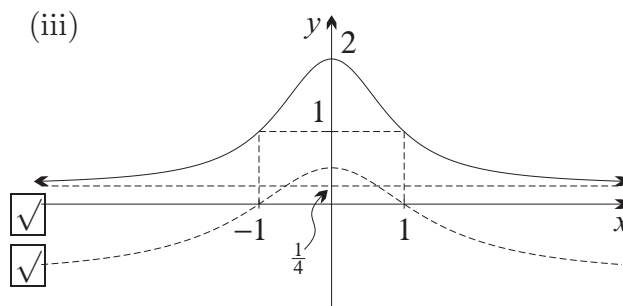


Shape at x-intercepts

y-intercept and horizontal asymptote



(iii)



$(-1, 1)$ and $(1, 1)$

y-intercept and horizontal asymptote



(b) (i) $(5, 0), (-5, 0), (0, 4)$ and $(0, -4)$



(ii) From $b^2 = a^2(1 - e^2)$

$$16 = 25(1 - e^2)$$

$$e^2 = \frac{9}{25}$$

so

$$e = \frac{3}{5}$$



(iii) $(3, 0)$ and $(-3, 0)$



(iv) $x = \frac{25}{3}$ and $x = -\frac{25}{3}$

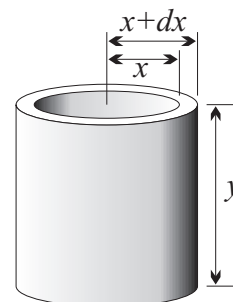


(c) The volume of the element is the difference between two cylinders, thus

$$dV = \pi(x + dx)^2 y - \pi x^2 y$$

$$= \pi y(2x + dx)dx$$

Sum the elements and take the limit as $dx \rightarrow 0$ to get



$$V = \int_0^1 2\pi xy \, dx$$



$$= 2\pi \int_0^1 \frac{x^2}{x^2 + 1} \, dx$$



$$= 2\pi \int_0^1 1 - \frac{1}{x^2 + 1} \, dx$$



$$= 2\pi \left[x - \tan^{-1} x \right]_0^1$$



$$= 2\pi - \frac{\pi^2}{2}$$



Total for Question 3: 15 Marks

QUESTION FOUR (15 marks)

(a) (i) $RHS = \sin A \cos B + \cos A \sin B$

$$\underline{-\sin A \cos B + \cos A \sin B}$$

$$= 2 \cos A \sin B$$



$$(ii) \quad \int_0^{\frac{\pi}{3}} 2 \cos 2x \sin x \, dx = \int_0^{\frac{\pi}{3}} \sin 3x - \sin x \, dx$$



$$= \left[-\frac{1}{3} \cos 3x + \cos x \right]_0^{\frac{\pi}{3}}$$

$$= \left(\frac{1}{3} + \frac{1}{2} \right) - \left(-\frac{1}{3} + 1 \right)$$

$$= \frac{1}{6}$$



$$\begin{aligned}
 \text{(b) (i)} \quad I_0 &= \int_0^1 \frac{1}{\sqrt{1+x}} dx \\
 &= 2 \left[\sqrt{1+x} \right]_0^1 \\
 &= 2\sqrt{2} - 2.
 \end{aligned}$$



$$\begin{aligned}
 \text{(ii)} \quad LHS &= \int_0^1 \frac{x^{n-1}}{\sqrt{1+x}} + \frac{x^n}{\sqrt{1+x}} dx \\
 &= \int_0^1 \frac{x^{n-1}(1+x)}{\sqrt{1+x}} dx \\
 &= \int_0^1 x^{n-1} \sqrt{1+x} dx \\
 &= RHS.
 \end{aligned}$$



$$\begin{aligned}
 \text{(iii)} \quad I_n &= \left[2x^n \sqrt{1+x} \right]_0^1 - 2n \int_0^1 x^{n-1} \sqrt{1+x} dx \quad (\text{by parts}) \\
 &= 2\sqrt{2} - 2n(I_{n-1} + I_n) \quad (\text{by part ii})
 \end{aligned}$$



$$\text{so } (2n+1)I_n = 2\sqrt{2} - 2nI_{n-1}$$



$$\text{or } I_n = \frac{2\sqrt{2} - 2nI_{n-1}}{2n+1}.$$

$$\begin{aligned}
 \text{(iv)} \quad I_1 &= \frac{1}{3} (2\sqrt{2} - 2I_0) \\
 &= \frac{1}{3} (4 - 2\sqrt{2}). \\
 I_2 &= \frac{1}{5} (2\sqrt{2} - 4I_1) \\
 &= \frac{1}{5} \left(2\sqrt{2} - \frac{16}{3} + \frac{8\sqrt{2}}{3} \right) \\
 &= \frac{1}{15} (14\sqrt{2} - 16).
 \end{aligned}$$



$$\text{(c) (i)} \quad (1+ic)^5 = 1 + 5ic - 10c^2 - 10ic^3 + 5c^4 + ic^5$$



$$\text{(ii)} \quad \text{Im}((1+ic)^5) = 0$$



$$\text{so } 5c - 10c^3 + c^5 = 0$$

$$\text{thus } c^4 - 10c^2 + 5 = 0 \quad (\text{since } c \neq 0)$$



(iii) The equation is a quadratic in c^2 , thus

$$c^2 = \frac{10 + \sqrt{80}}{2} \quad \text{or} \quad \frac{10 - \sqrt{80}}{2}$$



$$\text{hence } c = \sqrt{5 - 2\sqrt{5}}, -\sqrt{5 - 2\sqrt{5}}, \sqrt{5 + 2\sqrt{5}} \quad \text{or} \quad -\sqrt{5 + 2\sqrt{5}}.$$

$$\text{(iv)} \quad (r \operatorname{cis} \theta)^5 = r^5 \operatorname{cis} 5\theta \quad (\text{by de Moivre})$$

and since this is real

$$\sin 5\theta = 0$$

$$5\theta = n\pi$$

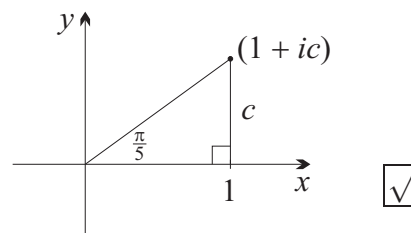
$$\text{or } \theta = \frac{n\pi}{5}$$



Thus the smallest positive value is $\theta = \frac{\pi}{5}$.

(v) This corresponds to the smallest positive value of c .

$$\begin{aligned}\text{Thus } \tan \frac{\pi}{5} &= \frac{c}{1} \\ &= \sqrt{5 - 2\sqrt{5}}.\end{aligned}$$



Total for Question 4: 15 Marks

QUESTION FIVE (15 marks)

(a) Since $P(z)$ has real coefficients, it follows that $z = 1 + 2i$ is also a zero. ☑

Let the remaining zero be α , then summing the roots

$$\alpha + 2 = \frac{3}{2}$$

or $\alpha = -\frac{1}{2}$ ☑

Hence $P(z) = 2(z + \frac{1}{2})(z - (1 - 2i))(z - (1 + 2i))$. ☑

(b) (i) Let the roots be α , α and β , then by the sums and products of roots

$$2\alpha + \beta = 0 \quad (1)$$

$$\alpha^2 + 2\alpha\beta = -p \quad (2)$$

$$\alpha^2\beta = q \quad (3)$$
☑

From (1), equations (2) and (3) become

$$3\alpha^2 = p \quad (4)$$

$$2\alpha^3 = -q \quad (5)$$
☑

hence $4p^3 = 4 \times 27\alpha^6$ (from equation (4))

$$= 27 \times 4\alpha^6$$
☑

$$= 27q^2 \quad (\text{from equation (5).})$$

(ii) Re-writing the equation of the cubic

$$x^3 - 3x - y = 0$$

which has a double root at the y -coordinates of the stationary points, so

$$27y^2 = 4 \times 3^3 \quad (\text{from part (i)})$$

so $y^2 = 4$

thus $y = 2$ or -2 ☑

(c) (i) $|x| < 1$

or $-1 < x < 1$ ☑

for which $S = \frac{1}{1+x^2}$ ☑

(ii) Thus $\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$

so $\int \frac{dx}{1+x^2} = \int (1 - x^2 + x^4 - x^6 + \dots) dx$

and $\tan^{-1} x = (x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \dots) + C$ ✓✓

At $x = 0$, $\tan^{-1} 0 = 0$, so

$$C = 0$$
✓

thus $\tan^{-1} x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \dots$

(iii) $\tan \frac{\pi}{12} = \tan \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$ ✓

$$= \frac{\sqrt{3} - 1}{1 + \sqrt{3}}$$

$$= \frac{(\sqrt{3} - 1)^2}{2}$$
✓

$$= 2 - \sqrt{3}.$$

(iv) $\frac{\pi}{12} \doteq (2 - \sqrt{3}) - \frac{1}{3}(2 - \sqrt{3})^3 + \frac{1}{5}(2 - \sqrt{3})^5$

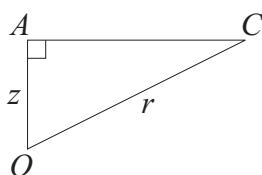
so $\pi \doteq 12 \left((2 - \sqrt{3}) - \frac{1}{3}(2 - \sqrt{3})^3 + \frac{1}{5}(2 - \sqrt{3})^5 \right)$

$$\doteq 3.1418 \quad (\text{correct to four decimal places})$$
✓

Total for Question 5: 15 Marks

QUESTION SIX (15 marks)

(a) (i)



In $\triangle OAC$ $AC^2 = r^2 - z^2$ (by Pythagoras) ✓

hence $|ABCD| = \frac{1}{2}AC^2$ (a square is a rhombus) ✓
 $= \frac{1}{2}(r^2 - z^2).$

(ii) The volume of the thin slice with thickness dz is $dV = \frac{1}{2}(r^2 - z^2) dz$

Sum the elements and take the limit as $dz \rightarrow 0$ to get

$$V = \frac{1}{2} \int_0^r (r^2 - z^2) dz$$
✓

$$= \frac{1}{2} \left[r^2 z - \frac{1}{3} z^3 \right]_0^r$$
✓

$$= \frac{1}{2} \left(r^3 - \frac{1}{3} r^3 \right) - 0$$

$$= \frac{1}{3} r^3.$$
✓

(b) Since S lies on PQ it follows that

$$\text{gradient } PS = \text{gradient } QS$$

$$\text{thus } \frac{b \tan \theta}{a \sec \theta - ae} = \frac{b \tan \phi}{a \sec \phi - ae}$$

$$\text{or } \frac{\tan \theta}{\sec \theta - e} = \frac{\tan \phi}{\sec \phi - e}$$

$$\text{whence } \tan \theta \sec \phi - e \tan \theta = \tan \phi \sec \theta - e \tan \phi$$

$$\text{and } e(\tan \theta - \tan \phi) = \tan \theta \sec \phi - \tan \phi \sec \theta.$$

$$\begin{aligned} \text{So } e &= \frac{\tan \theta \sec \phi - \tan \phi \sec \theta}{\tan \theta - \tan \phi} \times \frac{\cos \theta \cos \phi}{\cos \theta \cos \phi} \\ &= \frac{\sin \theta - \sin \phi}{\sin \theta \cos \phi - \cos \theta \sin \phi} \\ &= \frac{\sin \theta - \sin \phi}{\sin(\theta - \phi)}. \end{aligned}$$

$$(c) \quad (i) \quad \angle AFP = \beta \quad (\text{Alternate angles, } PS \parallel QR.)$$

$$(ii) \quad \angle RSA = \angle RAG \quad (\text{angle in the alternate segment of circle } SAR) \\ = \alpha.$$

$$\angle SPA = \angle RSA \quad (\text{angle in the alternate segment of circle } PBAS) \\ = \alpha.$$

$$(iii) \quad \angle FAP = \gamma \quad (\text{angle sum of } \triangle FAP)$$

$$\angle PQA = \angle QRA \quad (\text{angle in the alternate segment of circle } RABQ) \\ = \gamma.$$

Thus $\angle FAP = \angle PQA$

Hence FG is tangent to the circle through APQ by the converse of the angles in the alternate segment theorem.

Total for Question 6: 15 Marks

QUESTION SEVEN (15 marks)(a) (i) If $k < r$ then there is no term in x^r , hence ${}^kC_r = 0$. ✓

$$\begin{aligned}
 \text{(ii)} \quad \sum_{k=0}^n {}^kC_r &= \sum_{k=0}^n ({}^{k+1}C_{r+1} - {}^kC_{r+1}) \\
 &= ({}^1C_{r+1} - {}^0C_{r+1}) + ({}^2C_{r+1} - {}^1C_{r+1}) + ({}^3C_{r+1} - {}^2C_{r+1}) \\
 &\quad + \dots + ({}^{n+1}C_{r+1} - {}^nC_{r+1}) \\
 &= {}^{n+1}C_{r+1} - {}^0C_{r+1} \quad (\text{since all other terms cancel}) \\
 &= {}^{n+1}C_{r+1} - 0 \quad (\text{by part (i)}) \\
 &= {}^{n+1}C_{r+1}
 \end{aligned}$$
✓

$$\begin{aligned}
 \text{(iii)} \quad \sum_{k=0}^n k &= \sum_{k=0}^n {}^kC_1 \\
 &= {}^{n+1}C_2 \quad (\text{by part (ii)}) \\
 &= \frac{1}{2}n(n+1).
 \end{aligned}$$
✓

$$\begin{aligned}
 \text{(iv)} \quad (\alpha) \quad 2 \times {}^kC_2 + {}^kC_1 &= k(k-1) + k \\
 &= k^2.
 \end{aligned}$$
✓

$$\begin{aligned}
 (\beta) \quad \sum_{k=0}^n k^2 &= \sum_{k=0}^n 2 \times {}^kC_2 + {}^kC_1 \\
 &= 2 \times {}^{n+1}C_3 + {}^{n+1}C_2 \quad (\text{by part (ii)}) \\
 &= \frac{1}{3}(n+1)n(n-1) + \frac{1}{2}(n+1)n \\
 &= \frac{1}{6}(n+1)n(2(n-1) + 3) \\
 &= \frac{1}{6}(n+1)n(2n+1).
 \end{aligned}$$
✓

(b) The focal length = $\frac{1}{4a}$ ✓

$$\begin{aligned}
 \text{At the vertex } y &= x(ax + b) \\
 &= -\frac{b}{2a} \left(-\frac{b}{2} + b\right) \\
 &= -\frac{b^2}{4a}
 \end{aligned}$$

hence the directrix has equation

$$\begin{aligned}
 y &= -\frac{b^2}{4a} - \frac{1}{4a} \\
 &= -\frac{b^2+1}{4a}.
 \end{aligned}$$
✓

$$\begin{aligned}
 \text{(c)} \quad \text{(i)} \quad y &= -\frac{\tan^2 \alpha + 1}{4 \left(\frac{-g \sec^2 \alpha}{2V^2} \right)} \\
 &= \frac{V^2 \sec^2 \alpha}{2g \sec^2 \alpha} \\
 &= \frac{V^2}{2g}
 \end{aligned}$$
✓

✓

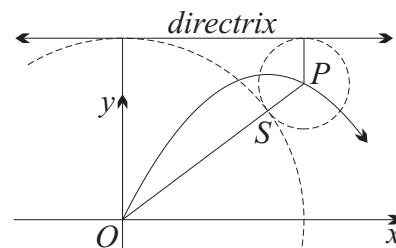
- (ii) The origin lies on the parabola so is equidistant from the focus and directrix. Thus there is a circle with centre the origin which passes through the focus and is tangent to the directrix.

Hence the radius of this circle is $\frac{V^2}{2g}$



and the equation is $x^2 + y^2 = \frac{V^4}{4g^2}$.

- (iii) Since P is on the parabola, P is equidistant from the focus and directrix. Hence there is a second circle with centre P which passes through the focus and is tangent to the directrix.



Since there is only one trajectory, the two circles intersect once only, at S .



Whence the two circles have a common tangent at S , which is perpendicular to both radii OS and SP . Hence O , S and P are collinear.



That is, OP is a focal chord.

Total for Question 7: 15 Marks

QUESTION EIGHT (15 marks)

(a) (i)
$$\begin{aligned} f(g) &= g - \frac{g^2}{g} - 2g \log \left(\frac{g}{g} \right) \\ &= g - g - 2g \log 1 \\ &= 0 \end{aligned}$$



(ii)
$$\begin{aligned} f'(x) &= 1 + \frac{g^2}{x^2} - \frac{2g}{x} \\ &= \left(1 - \frac{g}{x} \right)^2. \end{aligned}$$



- (iii) Now $f(g) = 0$
and $f'(x) > 0$ for $x \neq g$ (that is, $f(x)$ increasing for $x > g$)
hence $f(x) > 0$ for $x > g$.



(b) (i) $v' = -(g + kv)$

so $\frac{kv'}{g + kv} = -k$

Integrate with respect to t to get

$$\int \frac{kv'}{g + kv} dt = \int -k dt.$$



Note on the LHS that the numerator is the derivative of the denominator

so $\log(g + kv) = -kt + C_1$ (for some constant C_1)

or $g + kv = Ae^{-kt}$ where $A = e^{C_1}$.

At $t = 0$ $g + kv_0 = A$

so $g + kv = (g + kv_0)e^{-kt}$.



Integrate again with respect to t to get

$$gt + ky = -\frac{1}{k}(g + kv_0)e^{-kt} + \frac{1}{k}C_2 \quad (\text{for some constant } C_2)$$

so $kgt + k^2y = -(g + kv_0)e^{-kt} + C_2$.



At $t = 0$ $0 + 0 = -(g + kv_0) + C_2$

so $C_2 = (g + kv_0)$.



Thus $kgt + k^2y = (g + kv_0)(1 - e^{-kt})$

or $ky^2 = (g + kv_0)(1 - e^{-kt}) - gkt$.

(ii) At $t = T$, $v = 0$ so

$$g = (g + kv_0)e^{-kT}$$

or $e^{kT} = \frac{g + kv_0}{g}$

so $T = \frac{1}{k} \log \left(\frac{g + kv_0}{g} \right)$.



(iii) At $t = 2T$, $k^2y = (g + kv_0)(1 - e^{-2kT}) - 2gkT$

$$= (g + kv_0)(1 - (e^{-kT})^2) - 2gkT$$

$$= (g + kv_0) \left(1 - \left(\frac{g}{g + kv_0} \right)^2 \right) - 2g \log \left(\frac{g + kv_0}{g} \right)$$



$$= (g + kv_0) - \frac{g^2}{g + kv_0} - 2g \log \left(\frac{g + kv_0}{g} \right).$$

(iv) Let $x = g + kv_0$ then at $t = 2T$

$$k^2y = x - \frac{g^2}{x} - 2g \log \left(\frac{x}{g} \right)$$

$$= f(x)$$

$$> 0 \quad (\text{by part (a)})$$

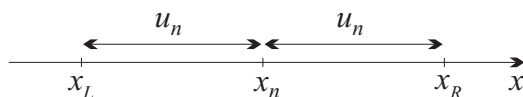


That is, it is above the ground and hence the downwards journey takes longer.

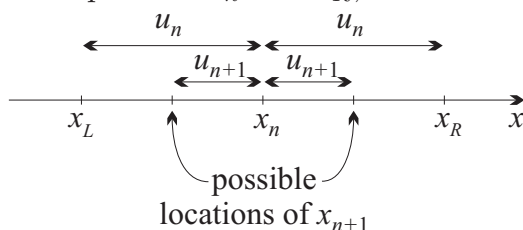
- (c) (i) The value x_n is the average of two numbers, one of which is x_{n-1} . Let these two numbers be x_L and x_R , where $x_L < x_n < x_R$. Thus

$$u_n = x_n - x_L = x_R - x_n.$$

The situation is shown on the number line.



Either x_{n+1} is the mid-point of x_L and x_n
or x_{n+1} is the mid-point of x_n and x_R , as shown in the diagram.



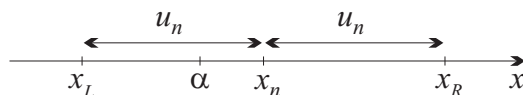
In either case the $|x_{n+1} - x_n| = \frac{1}{2}|x_n - x_{n-1}|$, viz $u_{n+1} = \frac{1}{2}u_n$. ✓

- (ii) Since $\frac{u_{n+1}}{u_n} = \frac{1}{2}$ for all n , u_n is a GP with common ratio $= \frac{1}{2}$. ✓

First term $u_1 = |b - a|$
 $= (b - a)$ ✓

Hence $u_n = (b - a) \left(\frac{1}{2}\right)^{n-1}$

- (iii) The root α lies between x_L and x_R , as in the diagram below.



Hence the distance from α to x_n is less than or equal to u_n . ✓

That is $|\alpha - x_n| \leq u_n$.

$$\begin{aligned} \text{(iv)} \quad \lim_{n \rightarrow \infty} |\alpha - x_n| &\leq \lim_{n \rightarrow \infty} u_n \\ &\leq \lim_{n \rightarrow \infty} (b - a) \left(\frac{1}{2}\right)^{n-1} \\ &\leq 0 \end{aligned}$$

hence $\lim_{n \rightarrow \infty} |\alpha - x_n| = 0$ ✓

thus $\lim_{n \rightarrow \infty} x_n = \alpha$.

Total for Question 8: 15 Marks